

CASE Tools



CASE (COMPUTER AIDED SOFTWARE ENGINEERING)

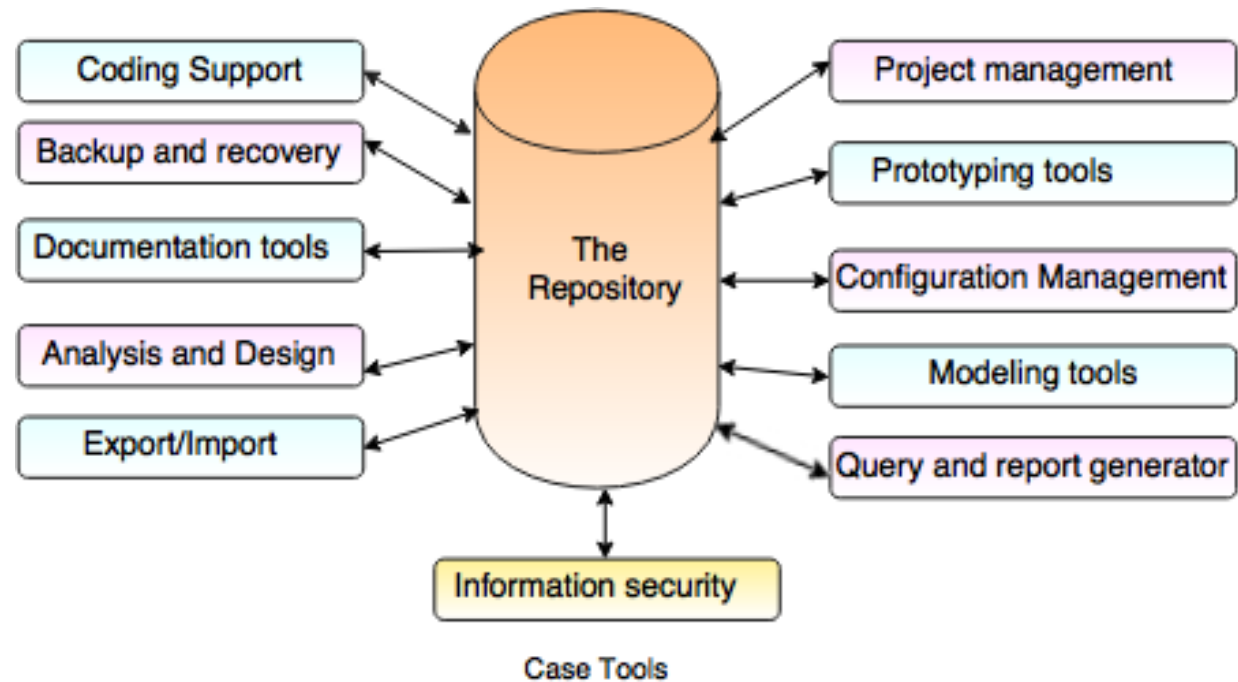
► Computer-Aided Software Engineering (CASE)

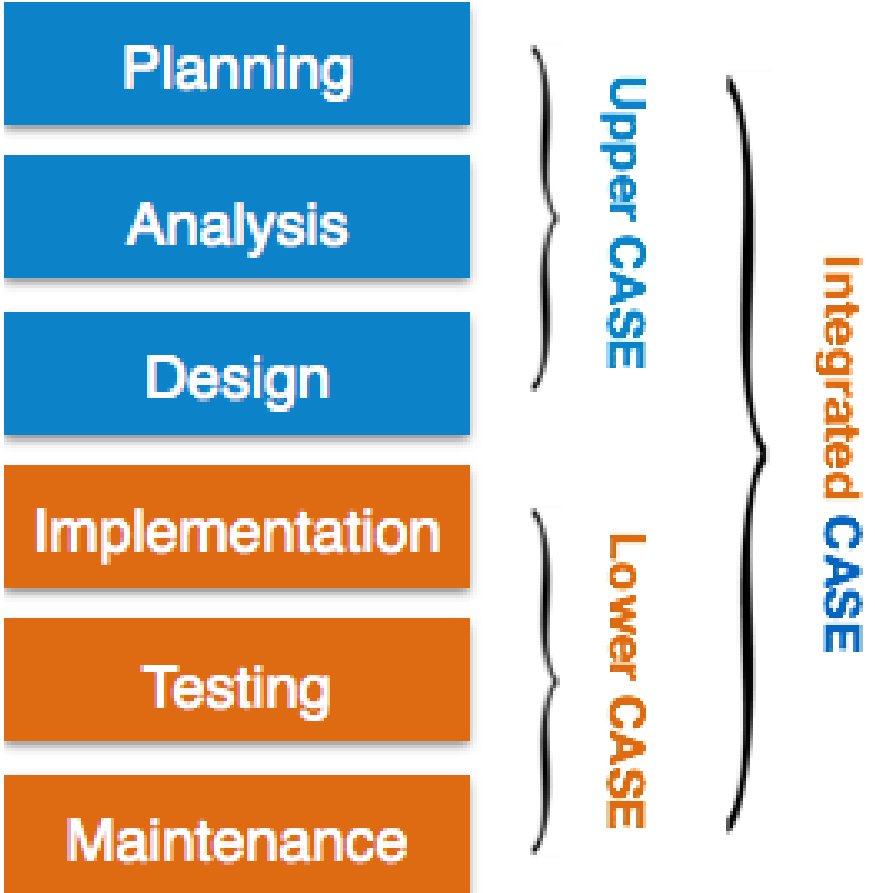
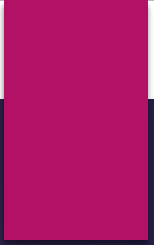
- **Computer-aided software engineering (CASE)** is the scientific application of a set of tools and methods to a software system which is meant to result in high-quality, defect-free, and maintainable software products. It also refers to methods for the development of information systems together with automated tools that can be used in the software development process.

Scope of Case Tools

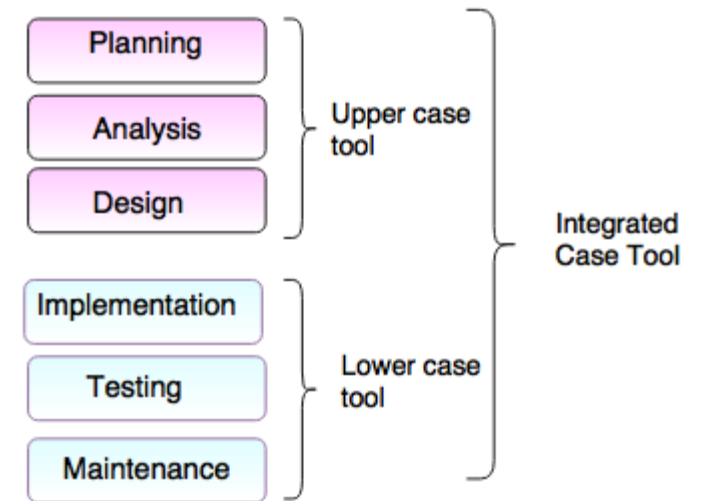
- ▶ The scope of CASE tools goes throughout the SDLC.

CASE Architecture





- **Upper Case Tools** - Upper CASE tools are used in planning, analysis and design stages of SDLC.
- **Lower Case Tools** - Lower CASE tools are used in implementation, testing and maintenance.
- **Integrated Case Tools** - Integrated CASE tools are helpful in all the stages of SDLC, from Requirement gathering to Testing and documentation.
- ▶ CASE tools can be grouped together if they have similar functionality, process activities and capability of getting integrated with other tools.



Case Tools Types

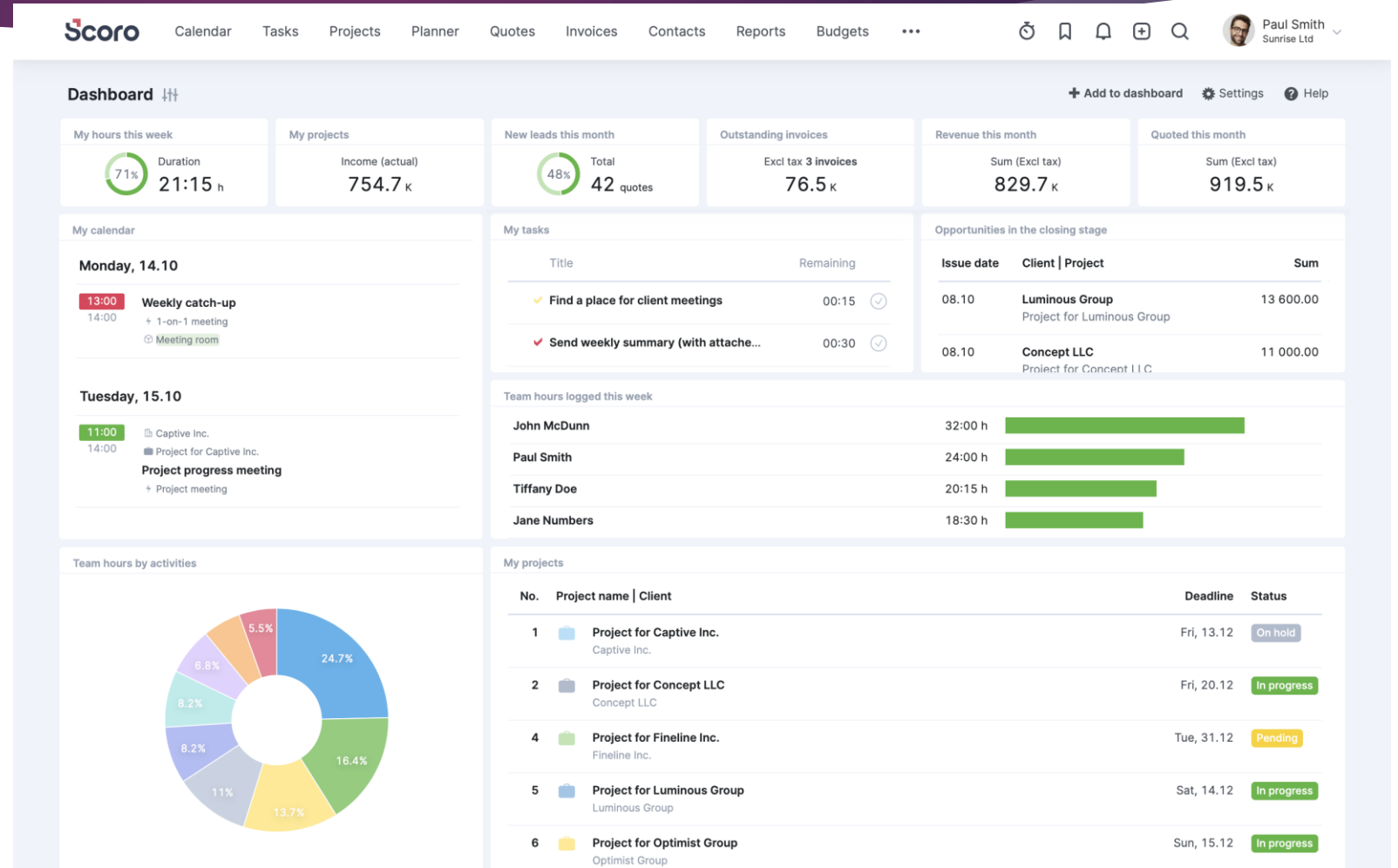
- ▶ **Diagram tools**
- ▶ These tools are used to represent system components, data and control flow among various software components and system structure in a graphical form. For example, Flow Chart Maker tool for creating state-of-the-art flowcharts.
- ▶ **Process Modeling Tools**
- ▶ Process modeling tools help the managers to choose a process model or modify it as per the requirement of software product. For example, GANTT CHART CREATORS

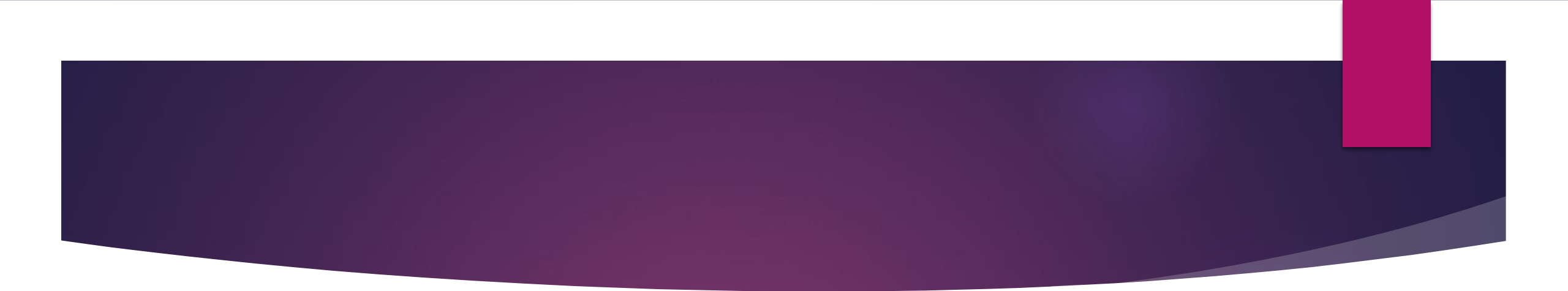
► Project Management Tools

1. Basecamp
2. SCORO
3. Workzone
4. Redmine
5. Teamweek
6. Gantt Pro
7. Ms project

► Documentation Tools

1. MS Office Suite
2. Document 360
3. Sublime
4. Visual studio Code



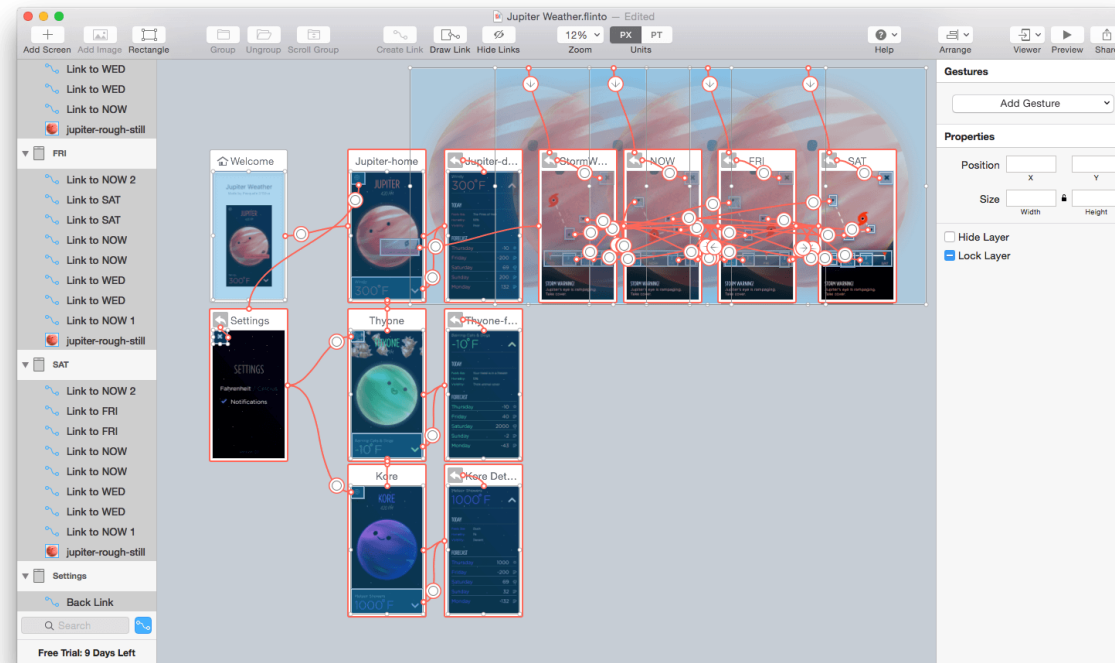


► Analysis Tools

1. Tableau
2. Knime
3. Orange
4. Microsoft BI

Design Tools

- ▶ 1. AXURE
- ▶ 2. ADOBE COMP
- ▶ 3. ADOBE XD
- ▶ 4. SKETCH
- ▶ 5. InVISion STUDIO
 - IBM Rational Rose
 - MS Visio
 - Argo UML
 - Magic Draw
 - Rational Software Architect (IBM)



► Configuration Management Tools

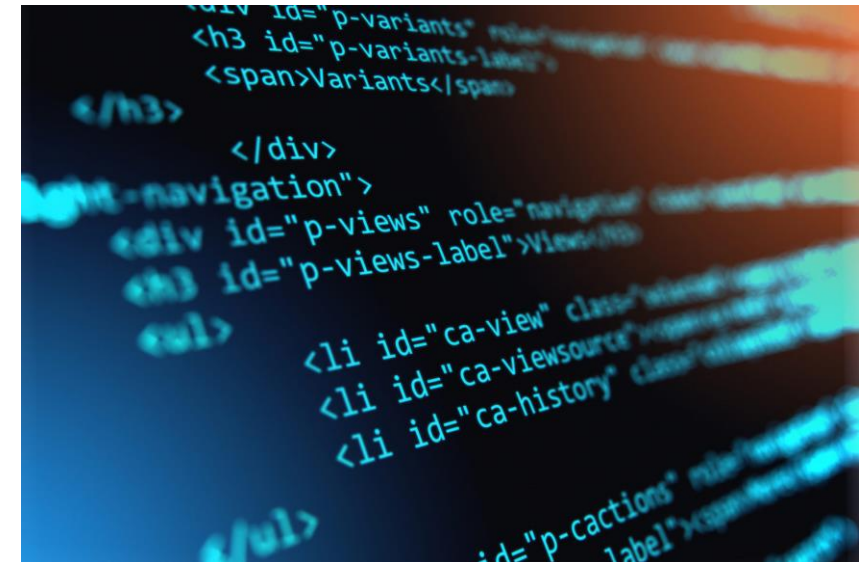
► Configuration Management tools deal with –

- Version and revision management
- Baseline configuration management
- Change control management

► CASE tools help in this by automatic tracking, version management and release management. For example, **Fossil, Git, Accu REV.**

Programming Tools

- ▶ These tools consist of programming environments like IDE (Integrated Development Environment), in-built modules library and simulation tools.
 - ▶ These tools provide comprehensive aid in building software product and include features for simulation and testing.
1. Eclipse /NETBEANS for java
 2. ANACONDA /IDLE
 3. Adobe DREAMWEAVER for HTML
 4. Microsoft Visual Studio
 5. CODE::BLOCKS, dev c++



TESTING TOOLS

- QTP/UFT
- RFT
- Test Manager
- Selenium
- Concordian
- JMEter
- Jprofiler
- Jprobe
- Loadrunner

Defect Tracking Tools

- JIRA
- ClearQuest
- Rally
- BugZilla

Agile Project Management Tools

- Rally
- Jira Agile
- Redmine
- Extreme Planner
- VersionOne
- TargetProcess
- AgileTrack
- Bamboo

Web server

- Apache Web Server (65% applications are hosted on apache)
- Internet Information Server (IIS) - 15% applications are hosted on IIS.
- Nginx from Nginx hosts around 12% of applications.
- GWS from Google (5%)
- Zeus Web Server
- iPlanet Web Server
- Roxen Web Server
- Jigsaw
- JRun
- Sambar Server
- Sun Java System Web Server